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Do Uncertainty and Financial Development Influence the FDI Inflow of a Developing Nation? A Time Series ARDL Approach

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Abstract: The study focuses on investigating the long-term and the short-term effect of uncertainty, and financial development on the FDI inflow of Pakistan during the period 2001–2019. To achieve the objective of this study, we obtained the data from World Development Indicators (WDI) and the European policy uncertainty index's websites. The dependent variable was FDI inflow. Experimental variables of the study are uncertainty and financial development. The stationarity testing revealed that FDI and Economic Policy Uncertainty (EUP) have weak significance and FD has no significance. However, by taking the first difference, all the variables become highly significant. Similarly, it is further indicated that the optimal lag level is four. Additionally, the bound test confirmed that a long-term relationship (co-integration) existed between the variables of the study. The ARDL estimations conclude that uncertainty and financial development have long-run as well as short-run effects on FDI inflow for Pakistan during the period of study. The uncertainty plays a strong part in decreasing the FDI inflow, whereas financial development plays a strong part in enhancing the FDI inflow in Pakistan during the period of study.

Keywords: uncertainty; financial development; FDI inflow; time-series; ARDL

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1. Introduction

The amount of non-resident shareholders' inward foreign investment throughout the reporting economy, comprising reinvested earnings as well as intra-company debts, exclusive of capital outflow including debt payment, is referred to as FDI inflows [1]. Higher FDI benefits both the manufacturing and service sectors [2]. This, by consequence, not only creates jobs, and yet helps to reduce joblessness even among the country's educated youth, both qualified as well as unqualified labor. Earnings are higher when there is a lot of work, giving the populace more purchasing power [3]. Pakistan was indeed the fourth-largest recipient of FDI inflows during the first part of 2019, at USD 33 billion. Inflows of FDI of Pakistan fell by −39 percent during 2020 because of the COVID-19 [4].

FDI inflow as a percentage of GDP as shown in the trend line of Pakistan during 2000–2019 in Figure 1 indicates a decreasing pattern during 2001–2004. However, it moved upward during 2005. The figure further indicates that after 2005 until 2014, a decreasing trend in FDI inflow for the economy of Pakistan was witnessed. However, it started moving upward after 2014, which still persists. The past studies carefully analyzed and compared all these trends of FDI inflow in Pakistan [4–6].

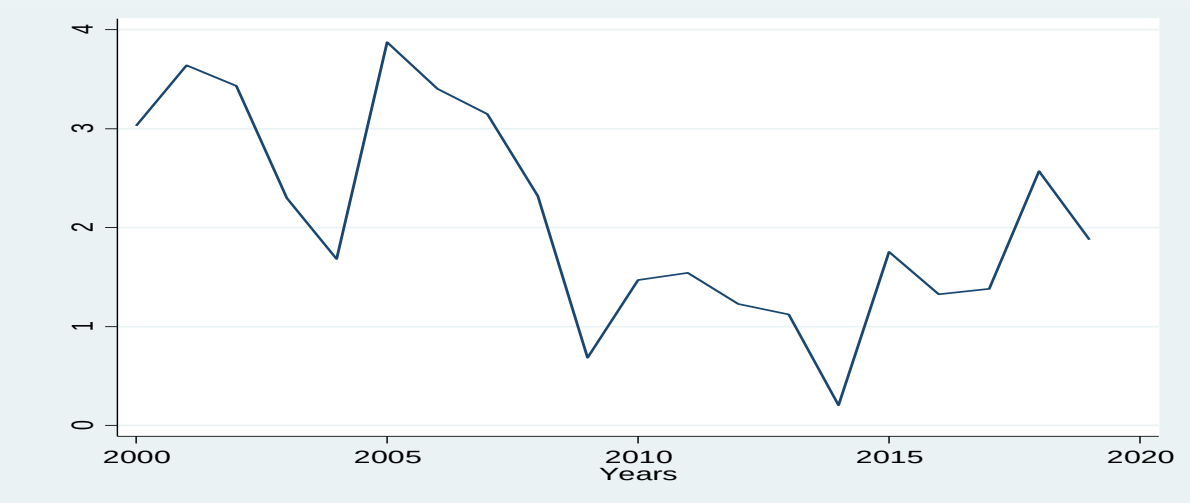


Figure 1. FDI inflow in Pakistan during 2001–2019 [7].

Financial firms, the financial industry, including financial products are all evolving in terms of financial development [8]. The development of financial activity volume as well as the procedure of financial sector upgrading has resulted in continual improvements in financial health [9]. It is evident that the removal of capital inflows and indeed the upgrading of the finance system, as well as financial product innovation as well as financial firms' diversification to react to financial activity contribute to the financial health [10]. The nation's economic prospects forecasts are unclearly caused by economic uncertainties [11]. When people focus on economic ambiguity, they generally mean there is a good chance that something negative may happen [12]. Figure 2 below indicates the upward trend-line for both uncertainties, as well as the financial development for Pakistan during 2000–2019. The increasing trends of uncertainty and financial development were comparable with the studies showing equivalent outcomes [4,13–17].

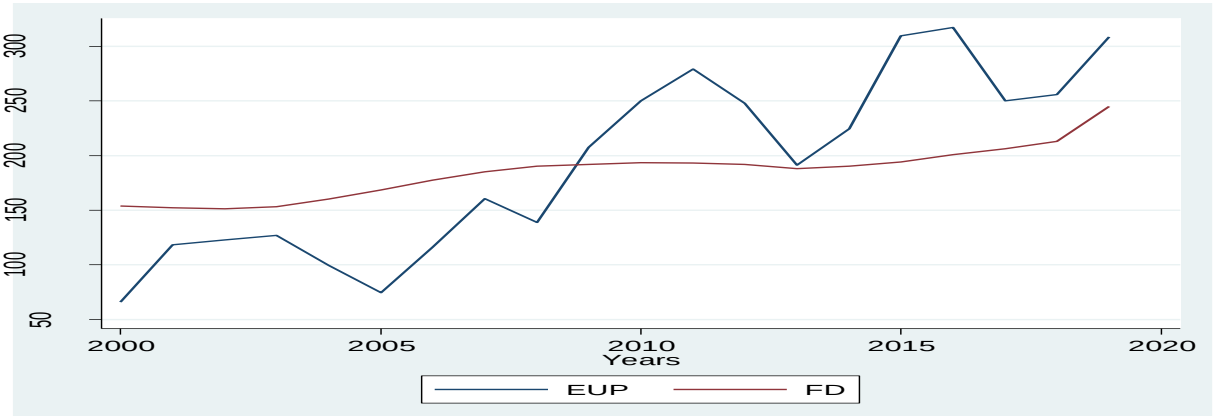


Figure 2. EUP and FD trend-line for Pakistan during 2000–2019 [18].

Ref. [19] stated that at first, the rate of increase in domestic EPU had a detrimental effect on FDI inflows. While regional policy uncertainty appears to have a detrimental influence on FDI inflows, research findings suggest that an increase in international (global) policy uncertainty might boost FDI inflows even more [12]. Likewise, the impact of financial development on determining FDI inflows need to be investigated [20].

The research mainly looked at the long-term as well as the short-term effects of uncertainty with financial development affecting Pakistan's FDI inflows across the observation period. The followings are indeed the study's core motivations:

- To investigate the influence of uncertainty affecting Pakistan's FDI inflows across the research period.
- To assess the impact of financial development on overall Pakistan's FDI inflow during the research period.

Researchers focused on the following research questions for viewing short- and long-term impact:

Research Question-1: How does uncertainty influence the FDI inflow of Pakistan during 2000–2020?

Research Question-2: What is the impact of financial development on the FDI inflow of Pakistan during 2000–2020?

2. Previous Research Investigations

The present study aims to explore the impact of uncertainty, and financial development on FDI inflow for Pakistan during 2000–2019. The critical literature analysis for this domain is as follows.

Ref. [21] conducted a study in Pakistan. The purpose of this research investigation was to find out the relationship between FDI and economic growth on a renewable energy context. Results show that chines FDI has potential effect on economic growth of Pakistan. Ref. [22] looked at Pak–China economic corridor investment and economic growth relationship.

This study used quarterly time series data from 2009 to 2018. In short-run there is no relationship between the chines economic corridor investment and economic development of Pakistan. Ref. [23] carried forward their investigation with regard to Turkey. The purpose of this study is to find out the relationship between the environment consequence of FDI influx and energy consumption. According to their findings, FDI long-run enhances environment consequences of Turkey. Ref. [24] looked at the dimension of political risk effect on economic and environmental development in Pakistan. Results represent that FDI, trade openness, BRI policy contribute to pollution and economic growth while political stability slows down the environment development but enhances economic development.

Endogenous growth theory stated that economic growth of a country depends on the country's domestic resources according to the national human development case of the economic development [25]. However, due to globalization, the economic growth of a country is now not solely dependent on the internal sources of a country. The multinational corporation FDI investment of a country changes the life of citizens by providing jobs to the local population which causes an uplift in the standard of living of the local community. This theory is in contrast with new classical growth model; according to this model the external factors such as technological advancement are the main source of economic growth [26].

Ref. [9] argued about a significant drop in FDI during as well as after the international economic meltdown; which is a period marked by increased uncertainty regarding economic plans in so many developed countries, including such non-conventional monetary as well as a fiscal policy such as the Brexit, but also global trade tensions. It indicated that short-term considerations, including policy uncertainty, could have an impact on FDI inflow. Similarly, a study forecasted a negative association between uncertainty and investment, with a solid reason to expect that such an effect is stronger with FDI than that for private investment [2]. For example, due to considerations linked with national borders, overseas investment has higher upfront costs unlike domestic investment [27]. Second, because international investors have little information about security from the guest country's financial and economic institutions, foreign direct investment is much more vulnerable to the political climate than private investment [28]. Given the promising importance of the relationship between FDI and policy uncertainty, limited research has looked at how these concepts relate as per the arguments [1,3,11,19,29–33].

Keeping in view the findings of previous research studies as well as the arguments presented in the past evidence in support of uncertainty and FDI inflow, the following

hypotheses are established for the long-run and short-run relationship testing for the abovesaid variables.

Hypothesis 1a (H1a). *The economic uncertainty policy should negatively affect the FDI inflow of Pakistan in the long term.*

Hypothesis 1b (H1b). *FDI inflow of Pakistan should decline by economic uncertainty policy in the short-run.*

Because of the transfer of technology embedded in FDI, as well as the massive amount of foreign capital involved, host markets are expected to profit from capital flows [3]. The infusion of foreign capital into the host nation may contribute toward the development of capital stock, while staff training may contribute toward the nation's professional development [8]. To put it another way, FDI can help any country to develop by increasing financial development—natural and institutional capital—or by increasing overall productivity levels [34]. According to the latest era of investigations, FDI inflows should be between various industries because multinationals want to avoid unauthorized access to prospective competitors yet gain from technology transfer to potential suppliers [35]. Limitations in these domains may limit domestic firms' ability to absorb emerging innovations and react to foreign players' risks and solutions [9]. Similar, positive relationships were discovered between financial development and FDI inflow [10,12,20,22,36,37].

Hypothesis 2a (H2a). *The financial development should positively affect the FDI inflow of Pakistan in the long term.*

Hypothesis 2b (H2b). *FDI inflow of Pakistan should enhance by financial development in the short-run.*

Figure 3 indicates the theoretical relationship between uncertainty and FDI inflow as well as between financial development and FDI inflow for Pakistan during the period 2000–2019. The theoretical framework of this study as well as the hypothesis and research questions are formed by considering the past literature.

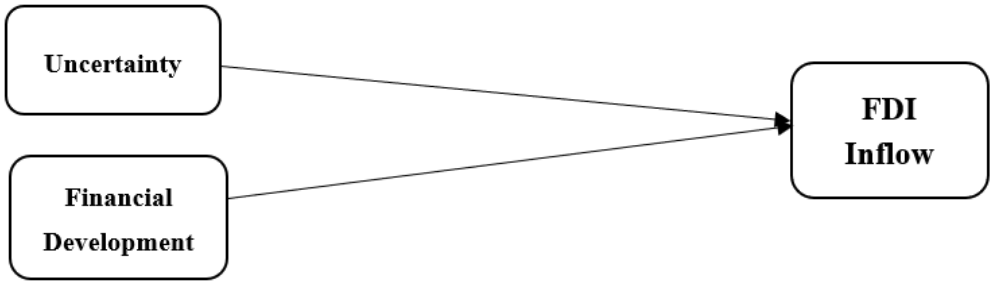


Figure 3. Theoretical framework.

3. Methodology

The present study meant to explore the effect of uncertainty and financial development on the FDI inflow of Pakistan for the period of 2001–2019. For achieving this objective, the data were obtained from WDI and the European policy uncertainty index's websites. The dependent variable was FDI inflow. The independent variables of the study include uncertainty and financial development.

The measurement of FDI inflow, uncertainty, and financial development are presented in Table 1 with their data sources and literature evidence.

Table 1. Measurements and sources of variables.

Variable	Variable Name	Measurement	Literature Source	Data Source
D.V	FDI inflow	FDI net inflow % GDP	[12,33,37]	WDI
IV1	Uncertainty	Economic Policy Uncertainty Index	[11,12,19,33]	Europe policy uncertainty data
IV2	Financial Development	It is the sum of (Monetary sector as a proportion of Private credit as % GDP) + (Domestic credit as a proportion of private credit by banks as % of GDP).	[12,20]	WDI

The study being a short-time series in nature with a fewer number of observations requires an appropriate method of estimation. A time-series analysis requires the appropriate model selection based on unit-root testing. When the estimation of a unit-root test confirms the stationarity at any level, the procedure for further estimation includes optimal lag length, co-integration test, and error-corrected version of ARDL regression.

The study required to analyze the time-series data to achieve the research objectives for examining the impact of uncertainty, financial development on the FDI inflow of Pakistan during the period of study. FDI inflow is the dependent variable, while the uncertainty and financial development are the independent variables of the study that form the following functional relationship.

$$FDI\ inflow = f\ (Uncertainty, financial\ development) \quad (1)$$

The basic time-series econometric model for this study is as follows:

$$FDI\ Inflow_t = \beta_0 + \beta_1 (Uncertainty)_t + \beta_2 (Financial\ Development)_t + \varepsilon_t \quad (2)$$

The independent variables such as uncertainty and financial development are showing larger values. To streamline the values of uncertainty, and financial development as par with the values of the dependent variable: FDI inflow, a transformation is required in terms of natural log. This transformation of taking the natural log values of independent variables established the following third regression equation in time-series analysis.

$$FDI\ Inflow_t = \beta_0 + \beta_1 Ln(Uncertainty)_t + \beta_2 Ln(Financial\ Development)_t + \varepsilon_t \quad (3)$$

The conversion of equation 3rd into fourth resulted in ARDL long-run estimation model.

$$FDI\ Inflow_t = \beta_0 + \beta_1 (FDI\ Inflow)_{t-1} + \beta_2 Ln(Uncertainty)_{t-1} + \beta_3 Ln(Financial\ Development)_{t-1} + \varepsilon_t \quad (4)$$

Similarly, a fifth equation formed to estimate the long-term relationship between the variables of the study as follows:

$$\Delta FDI\ Inflow_t = a_0 + \beta_1 (FDI\ Inflow)_{t-1} + \beta_2 Ln(Uncertainty)_{t-1} + \beta_2 Ln(Financial\ Development)_{t-1} + \omega EC_{t-1} + \varepsilon_t \quad (5)$$

4. Results and Discussion

The present study meant to investigate the effect of uncertainty, and financial development on FDI inflow for the economy of Pakistan, during 2000–2019. The data for this purpose were obtained from WDI and European policy uncertainty index websites. The dependent variable of the study was FDI inflow, and the independent variables were uncertainty and financial development. For obtaining the specific aims and answering the specific research questions of the study, the analysis required to apply the time series estimations, which include stationarity testing, optimal lag length, bound test, and ARDL estimations for short-term as well as long-term regression.

The descriptive statistic explains the summary of variables of the study in the form of mean, S.D, kurtosis, range, minimum, maximum, and count (observations). These

summarized statistics are reported in Table 2 below. The mean reports the average value of each variable of the study, and standard deviation reports the distance of observed value from their standard average value as per [38]. Similarly, the kurtosis, and skewness indicate the normality of variables. The variables are considered normal if their kurtosis value falls between ±3, and the skewness value falls between ±10 as per the guidelines provided by [39]. Further explanation reported below in Table 2.

Table 2. Descriptive statistics.

	FDI Inflow	EUP	FD
Mean	2.10	193.23	184.99
Standard Deviation	1.05	82.16	23.62
Kurtosis	−0.94	−1.39	0.80
Skewness	0.17	0.02	0.43
Range	3.67	251.28	93.80
Minimum	0.20	65.84	151.13
Maximum	3.88	317.12	244.92
Count	20	20	20

Table 2 above reports that FDI inflow in Pakistan counts on average as 2.10 with a standard deviation of 1.05. The minimum value for FDI inflow in this country is 0.20, while the maximum for the same is 3.88. However, the uncertainty as indicated by EUP scored on average as 193.23 with a standard deviation of 82.16 in Pakistan during the period of study. The minimum value for the same is 65.84, while the maximum value is 317.12. The financial development of this country reported on average as 184.99 with a standard deviation of 23.62. The minimum value is 151.13, and the maximum value is 244.92 for the same. The skewness and kurtosis of all the three variables, FDI inflow, EUP, and FD indicate the normality of data by meeting their respective threshold level.

A vital statistical feature of the time-series data (or, more accurately, the mechanism that generates it) is that it does not change with time, which is known as stationarity [40]. For testing the stationarity of the time-series dataset, some statistical procedures such as the ADF unit root test as suggested by [41], and the PP unit root test as suggested by [42] are used. The test results are reported in Table 3 for all the three variables, FDI inflow, EUP, and FD at the level as well as at the first difference. The null hypothesis indicates the stationarity of data. The rejection of which can confirm the stationarity. A further explanation of these values is reported in Table 3.

Table 3. Stationarity test for FDI inflow, economic policy uncertainty, financial development.

Variables	Level	
	ADF	PP
FDI Inflow	−2.272 *	−2.688 *
LnEUP	−3.210 *	−2.97 *
LnFD	−1.364	−1.431
	First Difference	
FDI Inflow	−5.135 ***	−5.135 ***
LnEUP	−4.075 ***	−4.075 **
LnFD	−3.434 **	−3.589 **

Note: Significant at * $p < 0.10$, ** $p < 0.05$ and *** $p < 0.10$.

Table 3 above reports the PP and ADF test statistics with test statistic values. The table indicates that FDI inflow and EUP are weakly significant at level, while the FD is not significant at level. However, by taking the first difference, all the variables become highly significant with p-values < 0.000, which rejected the null hypothesis and confirmed the stationarity of the data for this study. This confirmed the threshold level as indicated by [40–42].

The optimal selection of lag length has some important considerations for measuring the response of the dependent variable toward its independent variable with laps of time [43]. For a time-series with annual frequency, a maximum of 2–4 lags may be considered optimal as per the suggestions of [44]. The estimated values as per the criteria of AIC, HQIC, and SBIC are reported in Table 3 below.

Table 4 above indicated an optimal lag level of four as per the criterion of AIC, HQIC, and SBIC, highly significant with a degree of freedom 9.

Table 4. Selection of optimal lag. Selection-order criteria. Sample: 2004–2019. Number of obs = 16.

Lag	LL	LR	df	*p*	FPE	AIC	HQIC	SBIC
0	−2.71569				0.00041	0.714461	0.721879	0.859321
1	22.8219	51.075	9	0.000	0.000054	−1.35274	−1.32306	−0.773295
2	38.9692	32.295	9	0.000	0.000026	−2.24615	−2.19423	−1.23213
3	57.2603	36.582	9	0.000	0.000013	−3.40753	−3.33335	−1.95893
4	81.7288	48.937 *	9	0.000	6.6×10^{-6} *	−5.3411 *	−5.24467 *	−3.45792 *

Note: Significant at * $p < 0.10$, Endogenous: FDI, leup, lfd; Exogenous: _Cons.

Ref. [45] established to see if there was a long-term connection amongst variables; researchers used a bound-testing technique combined with ARDL modeling. It is the most recent method for examining the co-integration relationship between the variables of the study [46]). This test's null hypothesis implies that there is no long-term association. The denial of this might corroborate the original study elements' co-integration (long-term) association. [47] suggested that the f-value of test statistic > critical f-values of upper bound confirms the co-integration and rejects the null hypothesis. Stata software uses the Pesaran test for testing the co-integration in the ARDL process as indicated by [48]. He estimated critical f-value and test statistic f-value at different critical levels; 1%, 2.5%, 5%, and 10% reported below in Table 5.

Table 5. Co-integration (using ARDL bound test).

Critical F-Values K = 2		
C.L	LB	UB
1%	5.15	6.36
2.50%	4.41	5.52
5%	3.79	4.85
10%	3.17	4.14
F-test statistic = 12.306		

Table 5 indicates the f-test statistic value as 12.306, which exceeds that of the essential f value's upper threshold ranges at 1%, 2.5%, 5%, and 10%. It indicates that the H0 is rejected and establishes the existence of a long-term link (co-integration) here between the original study's variables.

The ARDL model is indeed an OLS-based approach that represents either non-stationary or mixed order with integrated time series. A linear adjustment can be used to generate a vibrant ECM using ARDL [49]. The error-corrected version of ARDL regression estimations reported below in Table 6 indicates the long-run as well as the short-run relationship of IVs and DV of the study. The complete interpretation and discussion of reported results are indicated in Table 6.

Table 6. ARDL (2, 3, and 2) regression estimates.

	D.fdi	Coef.	Std. Err.	t	$p > t$
Adjustment	FDI				
	L1.	−1.5717	0.301946	−5.21	0.002
Long-Run	Ln(EUP)	−1.61406	0.717281	−2.25	0.065
	Ln(FD)	3.360769	1.68402	2.24	0.257
Short-Run	FDI				
	LD.	0.44866	0.194201	2.31	0.06
	Ln(EUP)				
	D1.	−0.0647	0.960126	−0.07	0.948
	LD.	1.077841	0.82835	1.3	0.241
	L2D.	−1.54409	0.79451	−1.94	0.1
	Ln(FD)				
	D1.	9.69465	3.778205	3.43	0.203
	LD.	35.92974	10.77506	3.33	0.016
	Constant	−11.8717	17.31396	−0.69	0.519

Sample: 2004–2019; Number of obs = 16; R-squared = 0.91; dj R-squared = 0.78; Log likelihood = −2.7588681; Root MSE= 0.4695.

Table 6 reports the short-term link as well as the long-term link between the variables of the study for the economy of Pakistan during the period 2000–2019. The long-run relationship between economic uncertainty policy (EUP), and FDI inflow proved to be negatively significant [21]. Similarly, financial development proved to be positively significant during the long-run period of the study. The FDI inflow of Pakistan strongly declined by 1.61406 points if one point of economic policy uncertainty increased [24]. Due to this negative influence of EUP on FDI inflow during the long term, H1a is accepted. The negative relationship between EUP and FDI inflow in the long term is comparable with the findings of [12,19,31]. Additionally, the FDI inflow of Pakistan is strongly enhanced by 3.360769 points with an increase of one point in financial development. Due to this positive influence of financial development on the FDI inflow of Pakistan during the long term, H2a is accepted. The long-term positive link between financial development and FDI inflow is comparable with the studies such as [9,12,43].

In contrast, the economic uncertainty policy is weakly significant at lag 2 difference in case of short-run for decreasing the FDI inflow. This negative relationship of EUP and FDI inflow in Pakistan during the short-term accepted the H1b. The short-term negative link between EUP and FDI inflow is comparable with the findings of [11,21,22,33,50]. Similarly, a strongly positive influence of financial development is observed in the short-run for FDI inflow in Pakistan during the period of study. H2b was accepted due to this positive link between financial development and FDI inflow during the short-term in Pakistan. The positive link between financial development and FDI inflow in Pakistan is comparable with the similar outcomes of [4,12,34,35].

Table 7 reports the four most important diagnostic tests for testing the assumptions of the ARDL model; absence of multicollinearity, absence of heteroscedasticity, normality, and absence of stationarity at 2nd difference [51]. Table 7 shows the diagnostic test findings that were projected.

Table 7. Diagnostic test result (multicollinearity, heteroscedasticity normality, stationarity).

Tests	Criterion	Test-Statistic
RESET (Ramsey)	Prob > F	0.1059
Lagrange Multiplier (Breush-Pagan)	Prob > Chi-square	0.4765
Durbin Watson	d-stat (6,24)	1.0945
Normality Testing	Prob (Skewness)	0.4536
	Prob (Kurtosis)	0.5437

Table 7 above confirmed that all the assumptions of ARDL estimations are met and none is violated. There is no multicollinearity, no heteroscedasticity, no normality, no issue of functional form, and finally no stationarity at the second difference.

5. Conclusions

The purpose of this study was to look at the long-term and short-term effects of uncertainty, and financial development on the FDI inflow of Pakistan during the period 2001–2019. The data for this purpose were obtained from two sources: WDI and European policy uncertainty index websites. The dependent variable of this study was FDI inflow, and the independent variables were uncertainty and financial development. The estimations of the data require the time series analysis and due to the requirements of the study for testing the long-term as well as a short-term relationship, ARDL error-corrected estimations were executed along with unit-root testing (ADF, and PP), optimal lag length, co-integration testing (bound test), and ARDL diagnostic tests. The stationarity testing revealed that FDI inflow and EUP are weakly significant at a level, while the FD is not significant at a level. However, by taking the first difference, all the variables become highly significant with *p*-values < 0.000, which rejected the null hypothesis and confirmed the stationarity of the data for this study. Similarly, it further indicated that an optimal lag level of four as per the criterion of AIC, HQIC, and SBIC was confirmed along with highly significant value and a degree of freedom 9. Additionally, the bound test indicated the f-test statistic value as 12.306, which is exceeding the limit of the threshold for upper bound values range F at 1%, 2.5%, 5%, and 10%. It rejects the null hypothesis and confirms that a long-term relationship (co-integration) existed between the variables of the study.

The ARDL estimations revealed that a long-run relationship between economic uncertainty policy (EUP), and FDI inflow proved to be negatively significant. Similarly, financial development proved to be positively significant during the long-run period of the study. However, the economic uncertainty policy is weakly significant at lag 2 difference in case of short-run for decreasing the FDI inflow. Similarly, a strongly positive influence of financial development was observed in the short-run for FDI inflow in Pakistan during the period of study. Finally, the diagnostic tests revealed that there was no multicollinearity, no heteroscedasticity, no normality, no issue of functional form, and finally no stationarity at second difference. It was concluded from the estimated outcomes of ARDL estimations that uncertainty and financial development have long-run and short-run effect on FDI inflow for Pakistan during the period of study. The uncertainty plays a strong part in decreasing the FDI inflow, whereas financial development plays a strong part in enhancing the FDI inflow in Pakistan during the period of study. The conclusive evidence has practical implications for the policymakers in Pakistan for analyzing their practices for economic uncertainty policy as an impact of their FDI inflow. The study suggested them to enhance their financial development to accelerate their FDI inflow. On the basis of the results, it is suggested that the policymakers of Pakistan should make economic policies which encourage the FDI inflow. The main discouragement of FDI inflow is tex. Government should create a policy which mitigate the regulation for inflow of FDI. FDI inflow decreases the burden on the government for job market, because of private sector boom.

6. Limitations, Future Recommendation

The findings of the study are generalizable and comparable with single-country time series analysis. However, cross-country analysis with comparative mode can also be applicable in time series analysis. Future research may include further variables to analyze their impact on FDI inflow in Pakistan. Oil prices, consumer goods prices, automobile industry, agriculture product import export variable can be further included to test the effect of inflow and outflow of FDI on financial development.

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Ashraf) and M.A. (Mahmaod Alrawad); writing—original draft preparation, A.L., W.A.W., M.A. (Maryam Ashraf) and M.A. (Mahmaod Alrawad); writing—review and editing W.A.W. and M.A. (Maryam Ashraf) All authors have read and agreed to the published version of the manuscript.

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